

Sequence Models I: RNNs, BPTT, LSTMs and GRUs

Remembering with a hidden state — and gating what to keep

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

From fixed windows to recurrent state

Where we are

Last lecture built a language model over a **fixed window**: predict the next token from the previous k .

$$p_{\theta}(x_{t+1} \mid x_{<t}) \approx p_{\theta}(x_{t+1} \mid x_{t-k+1:t})$$

This lecture keeps the same **task** — next-token prediction — but changes **how the model remembers the past**: from a fixed window of inputs to a **running hidden state**.

Recall: modeling a sequence

A language model factorizes the joint probability by the chain rule:

$$p(x_{1:T}) = \prod_{t=1}^T p(x_t \mid x_{<t})$$

and we train it by minimizing the negative log-likelihood — token-level cross-entropy:

$$\mathcal{L} = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_{<t})$$

each factor is one V -way classification: “which token comes next?”

The fixed-window MLP, and its limits

The MLP-LM concatenates the last k embeddings and runs one MLP:

fixed context — anything before x_{t-k} is invisible; long-range dependencies are lost.

params grow with k — $W_h \in \mathbb{R}^{H \times kd}$; doubling context doubles the layer.

No shared state across time. The same token at position 1 vs 2 hits **different** columns of W_h — computation is not reused, and the input length is frozen at k .

What we actually want

A better sequence model should:

- read **one token at a time**, for a sequence of **any length**;
- keep a **memory** h_t — a summary of everything seen so far, $x_{1:t}$;
- **predict** from that memory, $\hat{y}_t = g(h_t)$;
- **share** the same parameters at every step.

a running state updated by **one** function: $h_t = f_\theta(h_{t-1}, x_t)$

The recurrent idea

Replace the fixed window with a **state** that is updated at every step:

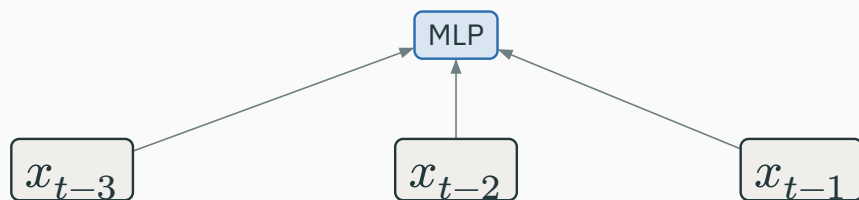
$$\underbrace{p(x_{t+1} \mid x_{t-k:t})}_{\text{MLP: fixed window}} \longrightarrow \underbrace{h_t = f_\theta(h_{t-1}, x_t)}_{\text{RNN: recurrent state}}$$

One **shared** function f_θ applied over and over gives us, for free:

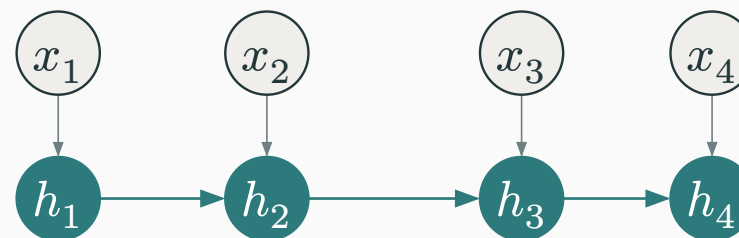
Property	Why it follows
variable length	same f_θ runs for as many steps as there are tokens
parameter sharing	one f_θ , not one weight block per position
learned state	h_t is a trainable summary of $x_{1:t}$
online	update as each token arrives — no need to see the whole sequence

Fixed window vs recurrent state v

window $k = 3$ only



fixed window (MLP-LM)



state carries all of the past

recurrent (RNN)

the MLP forgets past the window; the RNN **carries** the past forward in h_t

The vanilla (Elman) RNN

The equations **D**

At each step t : look up the token, update the state, read out a prediction.

$$e_t = E[x_t] \quad (\text{embedding lookup})$$

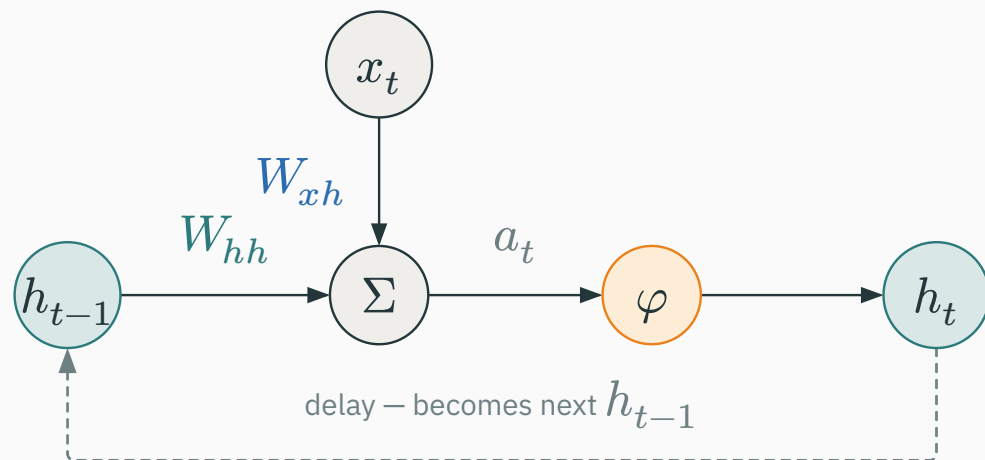
$$a_t = W_{xh} e_t + W_{hh} h_{t-1} + b_h, \quad h_t = \varphi(a_t)$$

$$z_t = W_{hy} h_t + b_y, \quad p_t = \text{softmax}(z_t)$$

$\varphi = \tanh$ usually; one set of weights W_{xh}, W_{hh}, W_{hy} reused at **every** step (Elman)

The RNN cell **v**

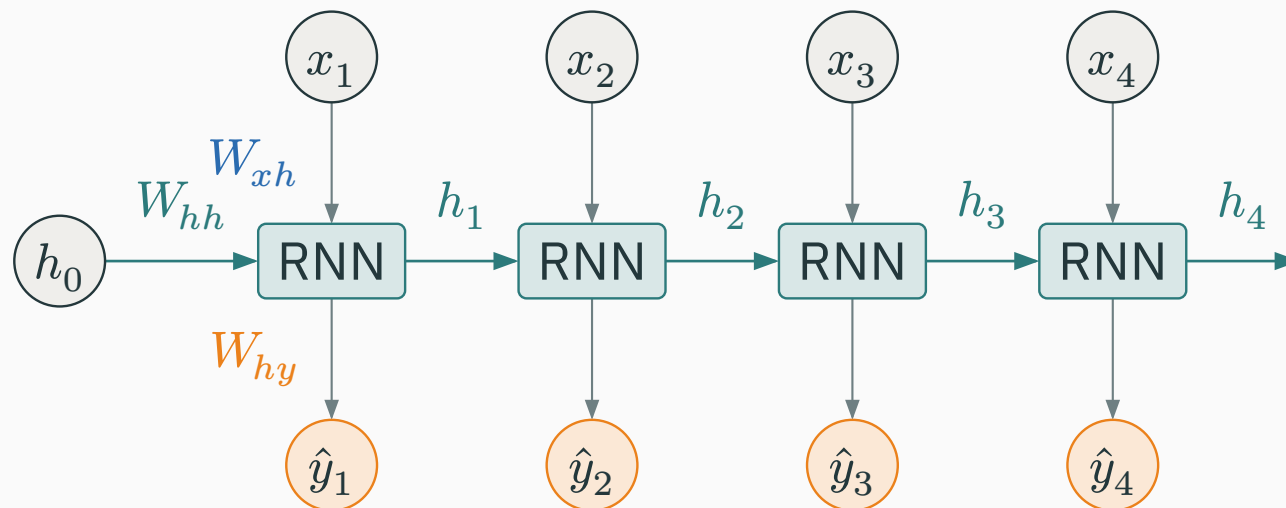
One shared cell: the past state h_{t-1} and the new input x_t meet, then φ gives h_t .



$$h_t = \varphi(W_{xh}e_t + W_{hh}h_{t-1} + b_h) - \text{the dashed edge is the **recurrence**}$$

Unrolling the RNN in time **v**

The **same** cell, drawn once per step — this is what backprop actually sees:



W_{xh} , W_{hh} , W_{hy} are **shared** across all steps — the diagram repeats one cell

The hidden state is memory

Unroll the recurrence and the whole past is nested inside h_t :

$$h_t = f(h_{t-1}, x_t) = f(f(h_{t-2}, x_{t-1}), x_t) = \dots$$

$$h_t = f(f(\dots f(h_0, x_1) \dots, x_{t-1}), x_t)$$

h_t is a **learned, fixed-size summary** of the entire prefix $x_{1:t}$ — no matter how long. The window is gone; the state remembers.

RNN for language modeling

Feed a sentence in; the **target at each step is the next token** (teacher forcing):

step t	1	2	3	4
input x_t	deep	learning	is	fun
state	h_1	h_2	h_3	h_4
target x_{t+1}	learning	is	fun	<eos>

$\text{loss} = \sum_t \text{cross-entropy}(p_t, x_{t+1})$ – every position supervises the step before it

RNN task patterns

The same cell, wired differently, covers most sequence tasks:

Pattern	Shape	Example
many-to-one	$x_{1:T} \rightarrow y$	sentiment: read all, predict at h_T
many-to-many (aligned)	$x_{1:T} \rightarrow y_{1:T}$	POS tag each token
language model	$x_{1:T} \rightarrow x_{2:T+1}$	next-token at every step
seq2seq (encoder-decoder)	$x_{1:T} \rightarrow y_{1:U}$	translation — different length

Tensor shapes

Track a batch of B sequences, length T , embedding d , hidden m , vocab V :

Tensor	Shape	Meaning
token ids X	$B \times T$	one id per position
embeddings $E[X]$	$B \times T \times d$	one d -vector per token
hidden states H	$B \times T \times m$	the state at every step
logits Z	$B \times T \times V$	a next-token score at every step

time is now an **axis**, not a fixed slot — the model handles any T

A fully-worked RNN forward pass

The tiny scalar RNN

Strip it to **one** hidden unit — everything is a scalar so we can do it by hand:

$$h_t = \tanh(w_x x_t + w_h h_{t-1} + b)$$

$$w_x = 1, \quad w_h = 0.5, \quad b = 0, \quad h_0 = 0, \quad x = (1, 2, -1)$$

three steps, three states — watch the memory build

Step 1

$$a_1 = w_x x_1 + w_h h_0 + b = 1 \cdot 1 + 0.5 \cdot 0 = 1$$

$$h_1 = \tanh(1) \approx 0.762$$

with no past state ($h_0 = 0$), the first step is just the input through $\tanh$

Step 2

Now the past enters through $w_h h_1$:

$$a_2 = w_x x_2 + w_h h_1 = 1 \cdot 2 + 0.5 \cdot 0.762 = 2.381$$

$$h_2 = \tanh(2.381) \approx 0.983$$

the state is being pushed toward the +1 saturation of tanh

Step 3

A negative input, but the state still remembers the positive past:

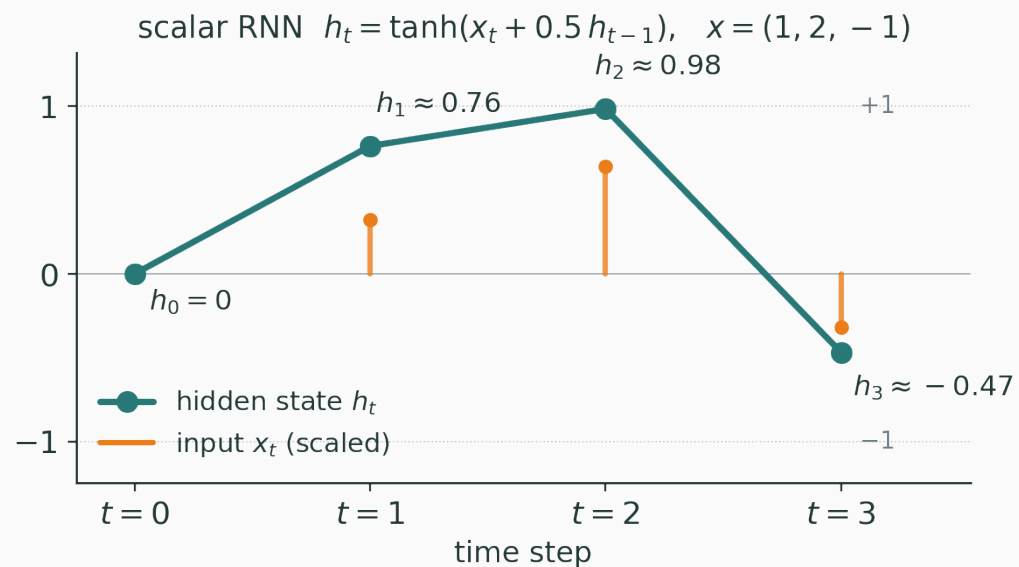
$$a_3 = w_x x_3 + w_h h_2 = 1 \cdot (-1) + 0.5 \cdot 0.983 = -0.509$$

$$h_3 = \tanh(-0.509) \approx -0.469$$

the $+0.49$ carried from h_2 **softened** the -1 input — that is memory at work

The forward trace v

t	x_t	$a_t = x_t + 0.5h_{t-1}$	$h_t = \tanh(a_t)$
1	+1	1.000	0.762
2	+2	2.381	0.983
3	-1	-0.509	-0.469



Why this is memory D

Substitute backwards

Why this is memory **D**

h_3 depends on **every** earlier input:

$$h_3 = \tanh \left(x_3 + 0.5 \underbrace{\tanh \left(x_2 + 0.5 \underbrace{\tanh(x_1)}_{h_1} \right)}_{h_2} \right)$$

x_1 reaches h_3 through **two** nested $\tanh$ and two factors of $w_h = 0.5$. Information persists — but each step **attenuates** it by w_h . Hold that thought (Part V).

Interactive: RNN forward **I**

INTERACTIVE

(to build) — set w_x, w_h, b and type a short sequence; watch h_t update step by step and the state trajectory trace out, exactly like the scalar example above.

Backpropagation through time

An RNN is a deep net in time

Unrolled, the RNN is just a **very deep** feedforward net whose layers **share weights**:

$$h_0 \rightarrow h_1 \rightarrow h_2 \rightarrow \dots \rightarrow h_T$$

training it = backprop on the unrolled graph = backpropagation through time (BPTT)

same chain rule as Lecture 3 — only now the “depth” is the sequence length T

The loss sums over time **D**

Every step contributes a term; the total loss is their sum:

$$\mathcal{L} = \sum_{t=1}^T \mathcal{L}_t, \quad \mathcal{L}_t = -\log p_{\theta}(x_{t+1} \mid x_{\leq t})$$

Because the weights are **shared**, each step's gradient lands on the **same** parameters:

$$\nabla_{W_{hh}} \mathcal{L} = \sum_{t=1}^T \nabla_{W_{hh}} \mathcal{L}_t$$

this is exactly Lecture 3's rule: a shared parameter **accumulates** gradient from every use

The backward recurrence

Write $\bar{u} = \partial \mathcal{L} / \partial u$. Walking one step back through the cell:

$$\bar{a}_t = \bar{h}_t \odot \varphi'(a_t) \quad (\text{through the tanh})$$

$$\nabla_{W_{xh}} += \bar{a}_t e_t^\top, \quad \nabla_{W_{hh}} += \bar{a}_t h_{t-1}^\top$$

$$\bar{h}_{t-1} += W_{hh}^\top \bar{a}_t \quad (\text{state gradient to the previous step})$$

Two sources of $\bar{h}_t$

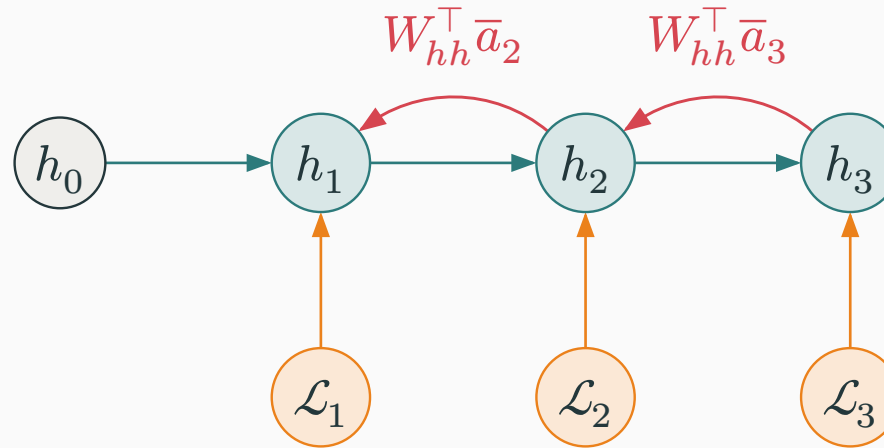
The state h_t is used **twice** — so its gradient is a **sum** of two paths:

$$\bar{h}_t = \underbrace{\frac{\partial \mathcal{L}_t}{\partial h_t}}_{\text{output path: its own loss}} + \underbrace{\frac{\partial \mathcal{L}_{\geq t+1}}{\partial h_t}}_{\text{future path: } W_{\text{top}}(\hat{h}_{t+1})}$$

Exactly the **branch rule**: h_t feeds both this step's readout **and** the next step's state, so gradients from both add. Miss the second term and long-range learning is impossible.

BPTT, unrolled **v**

Forward state (teal) flows right; the adjoint (orange = own loss, red = future) flows **back**:



each $\bar{h}_t$ collects its own loss ($\downarrow$) plus the future via $W_{hh}^\top$ ($\leftarrow$)

Truncated BPTT

Full BPTT over a length- 10^4 sequence is expensive and unstable. So **chunk** it:

- process the sequence in blocks of length K (say $K = 35$);
- backprop **within** a block; then **carry the state but detach it**:

$$h_{\text{start}}^{(\text{next block})} = \text{detach}\left(h_{\text{end}}^{(\text{this block})}\right)$$

the state keeps flowing forward; the **gradient** stops at the block boundary — bounded cost

INTERACTIVE

(to build) — step through the unrolled graph and watch each $\bar{h}_t$ accumulate its two contributions; toggle truncation length K and see where the gradient is cut.

Vanishing and exploding gradients

The long-range gradient is a product

To learn a dependency from step k to step T , the gradient runs the whole chain:

$$\frac{\partial h_T}{\partial h_k} = \prod_{j=k+1}^T \frac{\partial h_j}{\partial h_{j-1}}, \quad \frac{\partial h_j}{\partial h_{j-1}} = D_j W_{hh}$$

where $D_j = \text{diag}(\varphi'(a_j))$ is the activation slope at step j .

a product of $(T - k)$ matrices — it tends to **vanish or **explode****

The scalar recurrence lays it bare

Take one unit, $h_t = \tanh(w h_{t-1})$. Near 0, $\tanh'(0) = 1$, so:

$$\frac{\partial h_T}{\partial h_0} \approx w^T$$

$|w| < 1$: $w^T \rightarrow 0$ — **vanishing**. The distant past cannot send a gradient.

$|w| > 1$: $w^T \rightarrow \infty$ — **exploding**. Gradients blow up, training diverges.

Worked: w^{50}

Fifty steps back, with two nearby weights:

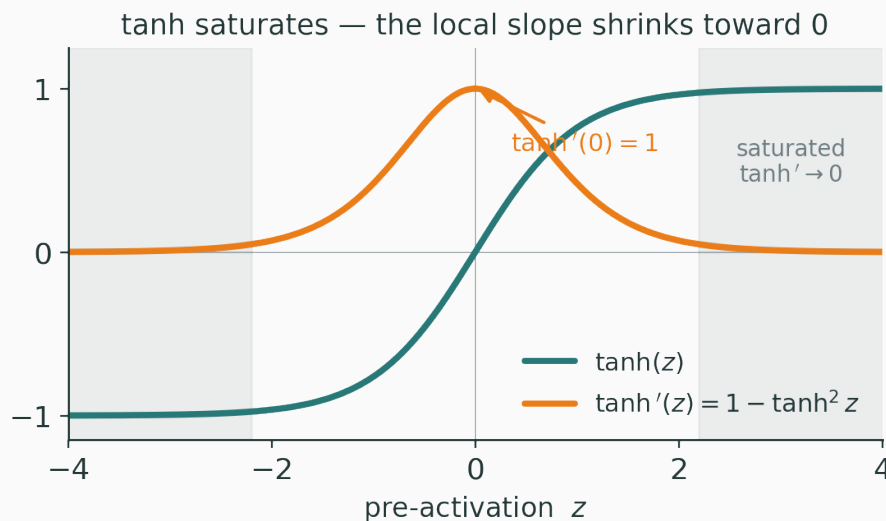
$$w = 0.9 : \quad 0.9^{50} \approx 0.005 \quad (\text{vanished})$$

$$w = 1.1 : \quad 1.1^{50} \approx 117 \quad (\text{exploded})$$

a 10% change in w flips the gradient from $\frac{1}{200}$ to $\times 117$ — over just 50 steps

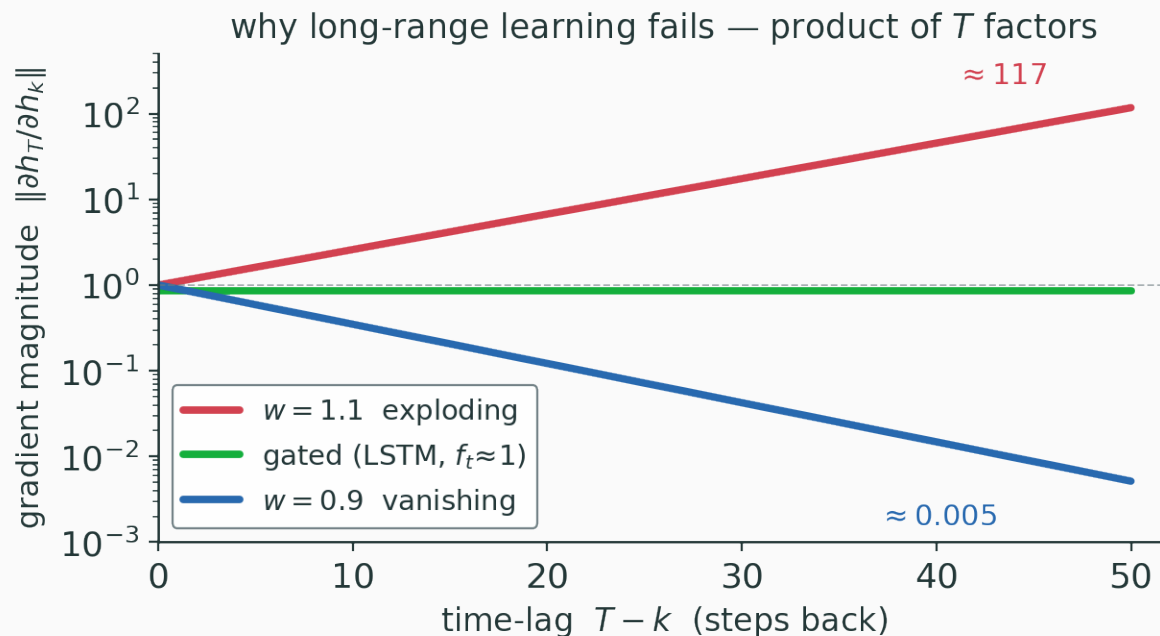
The saturation problem too **v**

$\tanh'(z) = 1 - \tanh^2(z) \leq 1$, and $\rightarrow 0$ once the unit saturates:



each $D_j \leq 1$ can only **shrink** the signal further — saturation makes vanishing worse

Gradient norm vs time-lag **v**



vanish (decay), explode (blow up), or — with a **gate** holding $f_t \approx 1$ — stay **flat**

Gradient clipping

Explosion has a cheap fix: cap the global gradient **norm** before the update.

$$\text{if } \|g\| > c : \quad g \leftarrow c \frac{g}{\|g\|}$$

Rescale, don't truncate — the **direction** of g is preserved, only its **length** is capped at c (Pascanu, Mikolov & Bengio 2013).

Worked: clipping

Threshold $c = 5$, gradient $g = (6, 8)$:

$$\|g\| = \sqrt{6^2 + 8^2} = 10 > 5$$

$$g \leftarrow 5 \cdot \frac{6, 8}{10} = (3, 4), \quad \|(3, 4)\| = 5 \quad \checkmark$$

same direction $(3, 4) \parallel (6, 8)$, length now exactly 5

Clipping is only half the cure

Fixes exploding — caps the step size so a single huge gradient cannot blow up training.

Does nothing for vanishing — you cannot rescale a gradient that has already decayed to ≈ 0 .

to carry gradients across long lags we must change the **architecture → gates**

Interactive: gradient dynamics I

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep the recurrent weight w and the lag T ; watch the long-range gradient collapse for $|w| < 1$, blow up for $|w| > 1$, and stay flat once a gate holds the memory open.

The LSTM

The core problem **D**

The vanilla RNN **overwrites** its state every step: $h_t = \varphi(W_{hh}h_{t-1} + \dots)$.

Every step multiplies the old state by W_{hh} and squashes it — memory is **rewritten**, and (Part V) the gradient is **multiplied** away. We want to **choose** what to keep.

let the network **decide**, per component: what to **forget**, what to **write**,
what to **expose**

Two states, one additive highway

The LSTM (Hochreiter & Schmidhuber, 1997) carries **two** vectors:

cell state c_t — the long-term memory (the “highway”).

hidden state h_t — the exposed output, a filtered view of c_t .

The cell updates by **addition**, not full overwrite:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

keep a gated fraction of the old memory, **add** a gated bit of new — this is the whole idea

Gates are learned $[0, 1]$ valves

A **gate** is a vector in $[0, 1]$, computed by a sigmoid from the input and previous state:

$$\text{gate} = \sigma(W u_t + b) \in (0, 1)^m, \quad u_t = [h_{t-1} ; x_t]$$

0 = block the channel, 1 = let it pass — applied **elementwise** with $\odot$

The LSTM equations

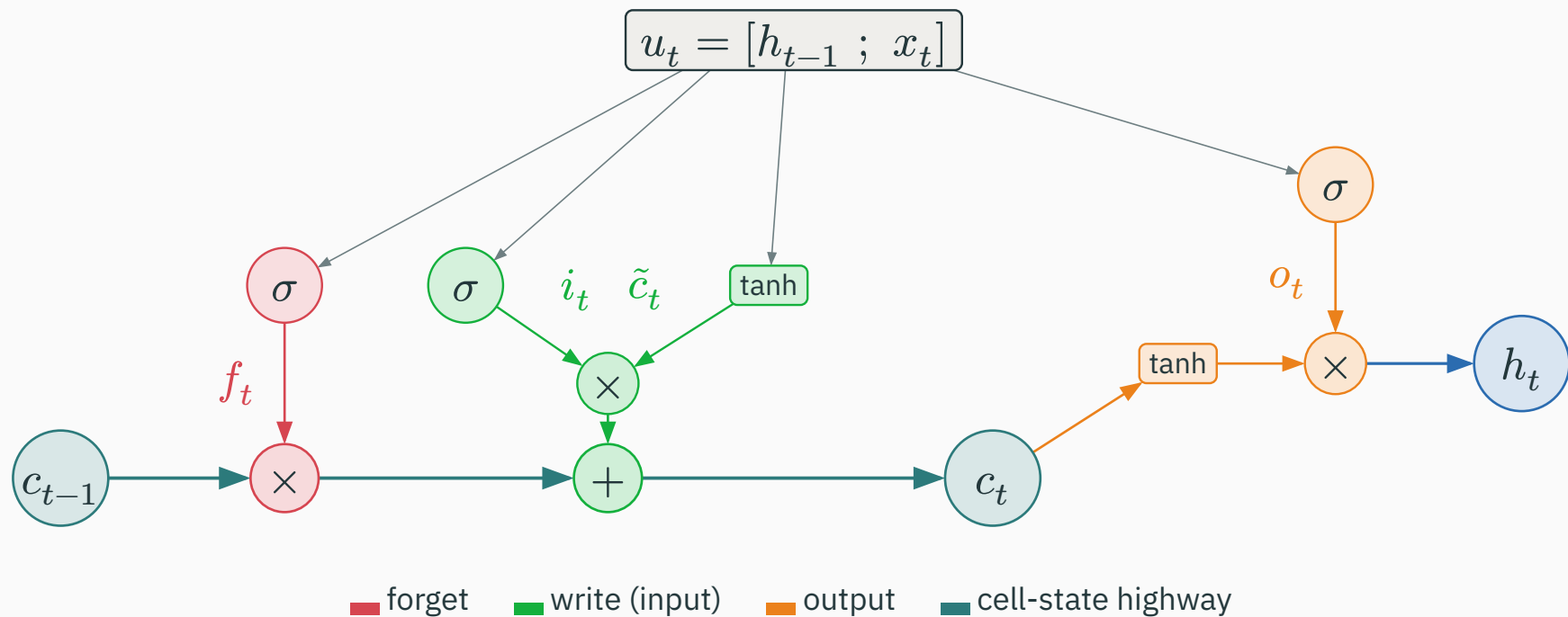
Shared input $u_t = [h_{t-1}; x_t]$; three sigmoid gates and one $\tanh$ candidate:

$$f_t = \sigma(W_f u_t + b_f), \quad i_t = \sigma(W_i u_t + b_i), \quad o_t = \sigma(W_o u_t + b_o)$$

$$\tilde{c}_t = \tanh(W_c u_t + b_c) \quad (\text{candidate memory})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t)$$

The LSTM cell v



cell state runs straight across the top; gates tap in to forget, write, and read out

Reading the three gates **D**

Gate	Operation	Meaning
forget f_t	$f_t \odot c_{t-1}$	1 = keep this memory, 0 = erase it
input i_t	$i_t \odot \tilde{c}_t$	how much of the new candidate to write
output o_t	$o_t \odot \tanh(c_t)$	how much of the memory to expose as h_t

candidate $\tilde{c}_t = \tanh(\dots)$ proposes **what** to write; the gates decide **how much**

Why additive memory helps

The cell state's step-to-step Jacobian is just the **forget gate** — no weight matrix:

$$\frac{\partial c_t}{\partial c_{t-1}} = \text{diag}(f_t)$$

So over many steps the gradient along the cell highway is a product of **gates**:

$$\frac{\partial c_T}{\partial c_k} = \prod_{j=k+1}^T \text{diag}(f_j)$$

if $f_j \approx 1$, the gradient flows **undamped — no repeated $\times W_{hh}$**

the central LSTM advantage: an **open** forget gate is a gradient highway across time

Worked scalar LSTM

One cell, old memory $c_{t-1} = 5$. **Retain** case — forget open, input nearly shut:

$$f_t = 0.8, i_t = 0.2, \tilde{c}_t = 3 : c_t = 0.8 \cdot 5 + 0.2 \cdot 3 = 4.6$$

Replace case — forget nearly shut, input open, candidate negative:

$$f_t = 0.1, i_t = 0.9, \tilde{c}_t = -2 : c_t = 0.1 \cdot 5 + 0.9 \cdot (-2) = -1.3$$

same equation: gates near $(1, 0)$ **hold** memory ($5 \rightarrow 4.6$); near $(0, 1)$ **overwrite** it ($5 \rightarrow -1.3$)

Interactive: the LSTM cell I

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/lstm-gates

Drag the forget, input and output gates and watch the cell state c_t retain, overwrite, or leak — and see the gradient highway open and close as f_t moves between 0 and 1.

The GRU

A simpler gated cell

The LSTM works but has **three** gates, a candidate, and **two** states. Can we do less?

The GRU (Cho et al. 2014, in the RNN encoder–decoder for translation) keeps the gating idea but uses **one** state and **two** gates.

The GRU equations

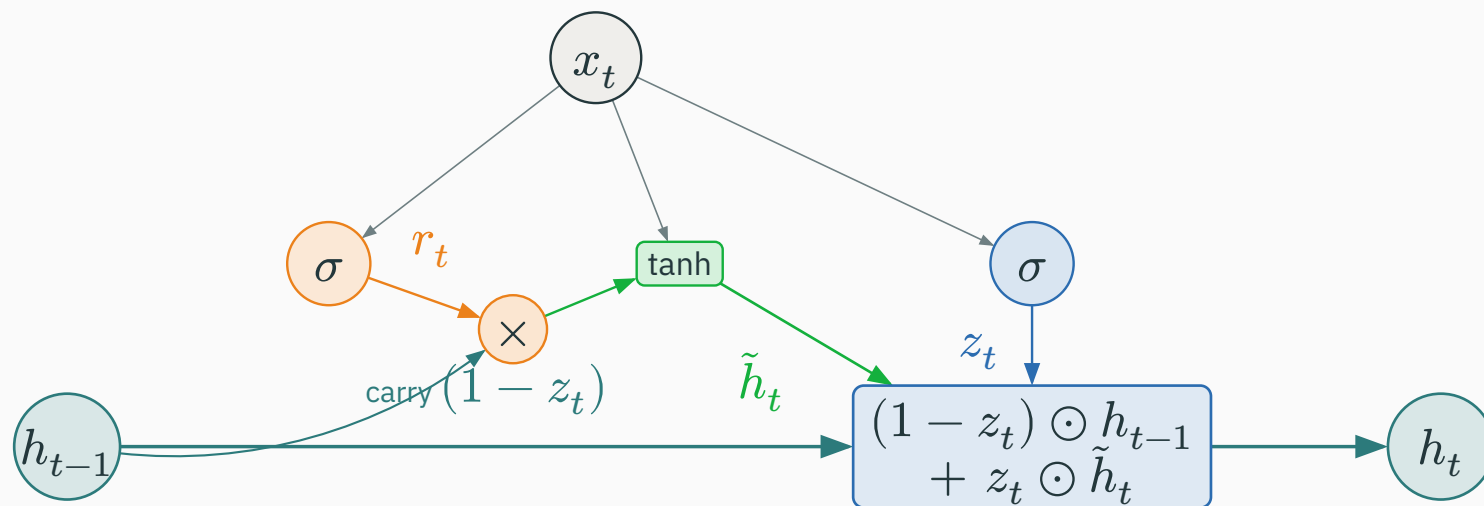
Update gate z_t , reset gate r_t , candidate $\tilde{h}_t$, then a **convex combination**:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z), \quad r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

The GRU cell v



reset r_t gates the past **into** the candidate; update z_t mixes old state and candidate

Reading the two gates D

update z_t : $h_t = (1 - z_t)h_{t-1} + z_t\tilde{h}_t$.
 $z = 0 \rightarrow$ copy old state; $z = 1 \rightarrow$ take candidate.

reset r_t : scales h_{t-1} **inside** the candidate.
 $r = 0 \rightarrow$ ignore the past, propose from x_t alone.

z_t plays the LSTM's forget + input as **one** knob; $z \approx 0$ is the gradient highway

Worked scalar GRU

One unit, old state $h_{t-1} = 4$, candidate $\tilde{h}_t = 0$. **Keep** — update nearly shut:

$$z_t = 0.25 : \quad h_t = 0.75 \cdot 4 + 0.25 \cdot 0 = 3$$

Replace — update wide open:

$$z_t = 0.9 : \quad h_t = 0.1 \cdot 4 + 0.9 \cdot 0 = 0.4$$

z small holds the state ($4 \rightarrow 3$); z large pulls it to the candidate ($4 \rightarrow 0.4$)

LSTM vs GRU v

	LSTM	GRU
states	two: h_t and c_t	one: h_t
gates	forget, input, output	update, reset
memory path	additive cell c_t , Jacobian $\text{diag}(f_t)$	convex mix, carry $(1 - z_t)$
parameters	more (4 weight blocks)	fewer (3 weight blocks)

both fix vanishing gradients; GRU is lighter, LSTM sometimes stronger — **try both, it is task-dependent**

Interactive: the GRU cell I

INTERACTIVE

(to build) — move the update and reset gates and watch the state interpolate between “keep the past” and “take the candidate”, side by side with the LSTM.

Sequence tasks with RNNs

Char-level RNN language model

Feed characters in, predict the **next** character at every step — the makemore idea, recurrent:

t	1	2	3	4
input	a	l	i	c
target	l	i	c	e

$p_t = \text{softmax}(W_o h_t + b_o)$; generate by **sampling** $\hat{x}_t \sim p_t$ and feeding it back

Sequence classification D

Many-to-one: read the whole sequence, predict **once**.

last state: $\hat{y} = \text{softmax}(Wh_T + b)$ —
the final h_T summarizes everything.

mean pool: $\hat{y} = \text{softmax}(W\bar{h} + b)$,
 $\bar{h} = \frac{1}{T} \sum_t h_t$ — robust to length.

sentiment, topic, spam — one label from a variable-length input

Bidirectional RNNs

For **non-causal** tasks, run two RNNs — forward and backward — and concatenate:

$$\vec{h}_t = \text{RNN}_{\rightarrow}(x_{1:t}), \quad \overleftarrow{h}_t = \text{RNN}_{\leftarrow}(x_{t:T}), \quad h_t = [\vec{h}_t ; \overleftarrow{h}_t]$$

Each h_t now sees the **whole** sequence — great for tagging or classification. But it peeks at the **future**, so it is **illegal for autoregressive generation**.

Padding and masking

Batches mix lengths — pad to a common T , then **mask the padding out of the loss**:

$$\mathcal{L} = \frac{\sum_{i,t} m_{it} (-\log p_{it})}{\sum_{i,t} m_{it}}, \quad m_{it} \in \{0, 1\}$$

$m_{it} = 1$ for real tokens, 0 for <pad>. The mask makes padding invisible to the gradient — otherwise the model learns to predict <pad>.

Training RNNs in practice

The training loop

Detach for truncated BPTT, mask the padding, clip the gradient:

```
h = model.init_hidden(batch_size)
for x, y, mask in loader:
    h = h.detach()                # truncated BPTT: cut the graph
    logits, h = model(x, h)       # forward one chunk
    loss = masked_cross_entropy(logits, y, mask)  # ignore <pad>
    optimizer.zero_grad()
    loss.backward()               # BPTT within the chunk
    nn.utils.clip_grad_norm_(model.parameters(), 5.0)  # tame explosions
    optimizer.step()
```

the three RNN-specific lines: detach, masked loss, clip_grad_norm_

Common bugs

Symptom	Likely cause
loss goes NaN	exploding gradients — clip; lower the learning rate
predicts <pad> a lot	no loss mask on padded positions
memory grows each step	state not detached — full graph kept forever
no long-term dependence	vanilla RNN vanishing — use LSTM / GRU
generation repeats itself	greedy decoding — sample / temperature / top- k
training is slow	not packing / batching variable lengths

RNN vs TCN — a preview

The RNN is inherently **sequential**: h_t needs h_{t-1} , so time cannot be parallelized.

RNN: $h_t = f(h_{t-1}, x_t)$ — a state carried step by step; $O(T)$ sequential.

TCN: stacked **dilated causal convolutions** — a fixed receptive field, fully **parallel** over time.

next lecture: convolutions over time, and then **attention** removes the recurrence entirely

Summary

One-slide summary v

Idea	Takeaway
recurrent state	$h_t = f_\theta(h_{t-1}, x_t)$ — shared f_θ , any length, memory in h_t
BPTT	unroll in time; shared-weight gradients accumulate over steps
vanilla RNN	$\partial h_T / \partial h_k \approx w^T$ — vanishing / exploding gradients
clipping	caps exploding norm; does nothing for vanishing
LSTM	cell $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$; forget / input / output gates
GRU	one state, update + reset gates; simpler, often as good

Final mental model — one idea

RNNs **remember** by carrying a hidden state.

LSTMs and GRUs **learn when** to remember, forget, and update.

a gate is a controllable path — for information **and** for gradients

We gave the model a **memory** and gates to control it.

Next: **attention and seq2seq** — let every step look **directly** at every other, no recurrence required.